STATISTICS · ANTICIPATING PATTERNS

Sampling Distribution Lab

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Sampling Distribution Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A population of wait times is strongly right-skewed with $\mu = 6.85$ minutes and $\sigma = 5.66$ minutes. You take many samples of size 50 and record $\bar{x}$ each time. Predict the shape, center and spread of the $\bar{x}$ values.
   Approximately normal, centered at 6.85 minutes, with SD $\sigma/\sqrt{50} \approx 0.80$ minutes (much less spread than the population).
2. What will the distribution of $\bar{x}$ look like if every sample has size n = 1?
   Just like the population (strongly right-skewed), because each $\bar{x}$ is a single value.

2 Explore use the tool

1. Skewed population, n = 1: draw 1,000 samples. Describe the shape, center and spread of the $\bar{x}$ values.
   Right-skewed like the population, centered near 6.85, SD near 5.66.
2. Increase n until the skewness of the sample means is below 0.35 (1,000+ samples). Record n and describe the shape.
   About n = 25 or more (theory: $n \geq 25$). The shape is close to normal (slight right skew).
3. Record the SD of $\bar{x}$ for n = 25 and for n = 100. What is the ratio? Why?
   About 1.13 and 0.57 ($\sigma/\sqrt{n}$). Ratio ≈ 2: four times the sample size halves the SD.
4. Bimodal population, n = 50: draw one sample. Is the sample normal? What is approximately normal?
   No, the sample looks bimodal like the population. Only the sampling distribution of $\bar{x}$ (many samples) is approximately normal.

3 Record use the tool and fill in the table

Sampling distributions

Draw 1,000 or more samples for each row. Simulated values vary a little; the key gives theory.

Population	n	Theory: SD of the statistic	Simulated SD	Shape
Skewed ($\sigma = 5.66$)	4	$5.66/\sqrt{4} \approx 2.83$	≈ 2.83	right-skewed
Skewed ($\sigma = 5.66$)	36	$5.66/\sqrt{36} \approx 0.94$	≈ 0.94	approximately normal
Uniform ($\sigma = 5.68$)	4	≈ 2.84	≈ 2.84	roughly normal (symmetric)
$\hat{p}$ with p = 0.1	20	$\sqrt{(0.1 \cdot 0.9/20)} \approx 0.067$	≈ 0.067	right-skewed (np = 2)
$\hat{p}$ with p = 0.1	100	$\sqrt{(0.1 \cdot 0.9/100)} = 0.03$	≈ 0.03	approximately normal (np = 10)

4 Explain what did you discover?

1. The mean of the sampling distribution of $\bar{x}$ is **μ**, so $\bar{x}$ is an **unbiased** estimator of μ.
2. $SD(\bar{x})$ = **$\sigma/\sqrt{n}$**. To cut it in half, multiply n by **4**.

3. Central Limit Theorem: for large n, the sampling distribution of $\bar{x}$ is approximately normal whatever the population's shape.

4. For $\hat{p}$, check Large Counts: np ≥ 10 and n(1 – p) ≥ 10.

5. A student says, "Because n = 50 is large, my sample is approximately normal." What is wrong with this statement?
   The CLT describes the sampling distribution of $\bar{x}$ (the means of many samples), not one sample. A single sample of 50 from a skewed population will itself look skewed, like the population.

5 Apply a new situation, no tool

1. Adult heights have μ = 66 in and σ = 4 in. For SRSs of size 16, find the mean and SD of $\bar{x}$.
   Mean 66 in; SD $4/\sqrt{16} = 1$ in.
2. Assuming heights are normal, find $P(\bar{x} > 67)$ for samples of size 16.
   $z = (67 - 66)/1 = 1$, so $P \approx 0.1587$.
3. A skewed population has σ = 12. What sample size makes $SD(\bar{x}) = 2$?
   $12/\sqrt{n} = 2$, so $n = 36$.
4. 8% of chips are defective. For samples of 150 chips, find the mean and SD of $\hat{p}$. Is a normal model reasonable?
   Mean 0.08; SD $\sqrt{(0.08 \cdot 0.92/150)} \approx 0.0222$. np = 12 and n(1 – p) = 138 are both ≥ 10, so yes.

Teacher notes

Intended discovery: The sampling distribution of a statistic is centered at the parameter (unbiased), its SD is $\sigma/\sqrt{n}$ or $\sqrt{(p(1-p)/n)}$, and for large n its shape is approximately normal regardless of the population (CLT); the Large Counts condition says when $\hat{p}$ is approximately normal.

Likely misconceptions:

- Confusing the distribution of one sample with the sampling distribution (Explore 4, Explain).
- Believing larger samples shift the center (they only shrink the spread).
- Thinking n ≥ 30 is a universal rule; strongly skewed populations need larger n.

Answers that can vary: All simulated values (Explore 1–3, Record "Simulated SD"); the n where skewness drops below 0.35 can vary slightly from run to run.